Operational Aftershock Forecasting: "oaf" Product Specification
Version 3

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This document describes the "`oaf`" product that is used to transmit Operational Aftershock Forecasts to PDL, which in turn forwards them to Comcat. It is assumed that the reader has a working knowledge of PDL and Comcat, which are documented at:

`https://ghsc.code-pages.usgs.gov/hazdev/pdl/index.html`
`https://earthquake.usgs.gov/data/comcat/`

Type

`oaf`	The product type for Operational Aftershock Forecasts.

Source

`us`	USGS Earthquake Science Center, Moffett Field, California. Note: The network code "`us`" refers to the USGS National Earthquake Information Center, however `oaf` products are actually produced by the USGS Earthquake Science Center.

Properties

`eventsource`	The seismic network for the mainshock. Usually, but not always, it is a 2-letter code.
`eventsourcecode`	The identification code for the mainshock, assigned by the seismic network. Note: The `eventsource` and `eventsourcecode` together form the mainshock's event ID. For example, if the event ID is "ci40925991" then the `eventsource` is "ci" and the `eventsourcecode` is "40925991".
`review-status`	The value is "`reviewed`" if the product was submitted manually by an analyst. This property is omitted if the product was submitted by the automatic system.

Contents

`forecast.json`	Machine-readable, JSON-formatted file that contains the forecast. The format of this file is documented below. Note: Prior to January 2022, this file did not exist. Instead, the JSON-formatted file was stored inline in the product, using a filename of "" (*i.e.*, the empty string). The correct way to access the forecast is to first check if `forecast.json` exists, and then, if it does not exist, look for the inline file.
`forecast_data.json`	Technical data and parameters used to compute the forecast. This file is primarily for assessing the performance of the OAF system, and its format is likely to change as the system evolves. Note: This file has a very complex format, and its content changes with different versions of the OAF software.

Format of `forecast.json`

The forecast is supplied as a JSON-formatted file. An example of a `forecast.json` file is appended at the end of this document. JSON is a simple file format for data interchange, which is documented here:

`https://www.json.org/json-en.html`

The `forecast.json` file contains a single JSON object, which contains several key/value pairs. Some of the values are themselves JSON objects or arrays, which means that key/value pairs can be nested several levels deep.

The top-level key/value pairs in `forecast.json` are described in the following table.

Table 1: Top-level key/value pairs.	
`creationTime`	The time at which the forecast was created. The value is an integer giving the number of milliseconds since midnight on January 1, 1970.
`expireTime`	The end time of the longest forecast contained in the file, which generally is one year in the future. The value is an integer giving the number of milliseconds since midnight on January 1, 1970.
`advisoryTimeFrame`	A text string which expresses the amount of time that the forecast advisory is in effect. It can be one of: "`1 Day`", "`1 Week`", "`1 Month`", or "`1 Year`". So far, the "`1 Day`" value has never been used. The event page uses this value to decide which forecast time periods to emphasize.
`template`	A text string which is used to select one of several layouts for the earthquake event page. For almost all forecasts it is "`Mainshock`". Other possibilities are "`Earthquake of Interest`" which is sometimes used for one-time manual forecasts, and "`Swarm`". The meaning and effect of assigning the value "`Swarm`" is not yet defined.
`injectableText`	A text string that is included in the Commentary section of the earthquake event page. An analyst can use this to provide additional information about the aftershock sequence. It is an empty string, "", if no injectable text is supplied.
`observations`	An array that summarizes the aftershocks that have been observed so far. Each element of the array is an object that contains the key/value pairs described in Table 2. Some of the observations appear in the Commentary section of the event page.

Table 1: Top-level key/value pairs.	
`model`	An object that contains a description of the statistical model used to construct the forecast. Table 3 describes the key/value pairs that appear in this object. This information is used to construct the Model Parameters section of the event page.
`fractileProbabilities`	An array that contains the fractiles used to construct the probability distribution for forecasted number of aftershocks. It is a sequence of floating-point numbers, in increasing order, greater than 0.0 and less than 1.0. The contents of this array can vary. Currently, it is most commonly an array that is 197 entries long and contains the values 0.01 through 0.99 in steps of 0.005. Note: This was added in April 2023.
`barLabels`	An array that contains the labels appearing in the bar graph in the Summary section of the event page. It is a sequence of integers, in increasing order. Currently, the array is 10 entries long, and contains the following values: 0, 1, 2, 5, 10, 20, 50, 100, 200, 500. Note: Currently, the bar graph consists of ten bars. The numbers 1 through 500 appear on tick marks separating the bars; the number 0 does not appear. The first bar represents 0 aftershocks; the second bar represents 1 aftershock; the third bar represents 2-4 aftershocks; the fourth bar represents 5-9 aftershocks; and so on. The last bar represents 500 or more aftershocks. Note: This was added in November 2023.

Table 1: Top-level key/value pairs.	
`forecast`	A table that gives the forecasted number of earthquakes for various time periods and magnitudes. At present, the time periods are 1 day, 1 week, 1 month (defined to be 30 days), and 1 year (defined to be 365 days). The magnitudes are almost always 3.0, 4.0, 5.0, 6.0, and 7.0; and larger than the mainshock. For very large earthquakes, it is possible for magnitude 8.0 to be included. For some small earthquakes, it is possible for magnitude 7.0 and possibly also magnitude 6.0 to be omitted. The value of `forecast` is an array. Each element of the array corresponds to one time period, and contains the forecasts for that time period, as described in Table 5. This information is used to construct the Summary and Forecast Table sections of the event page, plus some of the information in the Commentary section.
`nextForecastTime`	The time at which the next forecast is scheduled to occur. The value is an integer giving the number of milliseconds since midnight on January 1, 1970. The value is `-1` if no additional forecasts are scheduled. It is possible for `nextForecastTime` to be omitted. If it is omitted, it means that no information about the next forecast is available. Note: Prior to January 2022, this key/value pair was not included, and so `advisoryTimeFrame` was used to calculate the next update time that was displayed on the event page.

Each element of the `observations` array contains information about the aftershocks that have occurred so far, as described in the following table.

Table 2: Key/value pairs that appear within the `observations` array.	
`magnitude`	An aftershock magnitude. The magnitudes appearing in the `observations` array are almost always 3.0, 4.0, 5.0, 6.0, and 7.0. For very large earthquakes, it is possible for magnitude 8.0 to be included. For some small earthquakes, it is possible for magnitude 7.0 and possibly also magnitude 6.0 to be omitted.
`count`	The number of aftershocks observed so far, whose magnitude is at least as large as the value of `magnitude`.

The `model` object contains information about the statistical model used to create the forecast, as described in the following table.

Table 3: Key/value pairs that appear within the `model` object.	
`name`	A text string that identifies the statistical model used.
`reference`	A placeholder for the URL of a page that describes the statistical model. Currently the value is `"#url"` which indicates that no URL is provided.
`parameters`	The parameter values used in the statistical model. This is an object that contains a set of key/value pairs as shown in Table 4.

The `parameters` object contains parameter values for the statistical model. The parameters vary depending on the statistical model chosen. Also, this list changes over time, as the system evolves, and as more statistical models are introduced. Some parameters are optional, depending on the values of other parameters.

Table 4: Key/value pairs that appear within the `parameters` object.	
Parameters for the aftershock search region.	
`regionType`	The type of region used to collect aftershocks. It is either `"circle"` or `"rectangle"`. So far, all regions have been circular.
`regionCenterLat`	For a circular region, the latitude of the center of the circle, in degrees.
`regionCenterLon`	For a circular region, the longitude of the center of the circle, in degrees.
`regionRadius`	For a circular region, the radius of the circle, in kilometers.
`regionMinLat`	For a rectangular region, the latitude of the south side of the rectangle, in degrees.
`regionMaxLat`	For a rectangular region, the latitude of the north side of the rectangle, in degrees. Note: The value of `regionMaxLat` is always greater than the value of `regionMinLat`.
`regionMinLon`	For a rectangular region, the longitude of the west side of the rectangle, in degrees.
`regionMaxLon`	For a rectangular region, the longitude of the east side of the rectangle, in degrees. Note: The value of `regionMaxLon` is always greater than the value of `regionMinLon`. If the rectangle crosses the date line, then `regionMaxLon` is greater than 180. If the rectangle contains the north or south pole, then `regionMinLon` is -180 and `regionMaxLon` is 180.
Parameters for the mainshock.	
`magMain`	The mainshock magnitude.
Parameters for the Reasenberg-Jones statistical model. (see Reasenberg and Jones 1989, https://doi.org/10.1126/science.243.4895.1173)	
`a`	Reasenberg-Jones a-value, which is productivity. So far, this has always been a maximum likelihood estimate, but in principle it could be a fixed value.

Table 4: Key/value pairs that appear within the `parameters` object.	
`b`	Gutenberg-Richter b-value. This is a fixed value.
`p`	Omori exponent p. This is usually a fixed value, but sometimes it is a maximum likelihood estimate.
`c`	Omori time offset c, in days. This is usually a fixed value, but sometimes it is a maximum likelihood estimate.
`aSigma`	Standard deviation of the Reasenberg-Jones a-value. It is zero if a fixed a-value was used.
`pSigma`	Standard deviation of the Omori exponent p. It is zero if a fixed p-value was used. Note: There is no `cSigma` parameter, because the c-value is not well described by a Gaussian distribution.
Parameters for catalog incompleteness.	
`Mc`	Catalog magnitude of completeness. The `Mc` parameter is used in two situations: • When the magnitude of completeness is time-independent. • When the magnitude of completeness is time-dependent, but a Helmstetter model is not being used. Note that for large mainshocks, this may be larger than the configured value in order to limit the number of aftershocks processed.
`Mcat`	Normal catalog magnitude of completeness. The `Mcat` parameter is used when applying a Helmstetter model of time-dependent catalog incompleteness. Note that for large mainshocks, this may be larger than the configured value in order to limit the number of aftershocks processed.
`F`	F-parameter for a Helmstetter model of time-dependent magnitude of completeness. (see Helmstetter *et al*. 2006, https://doi.org/10.1785/0120050067)
`G`	G-parameter for a Helmstetter model of time-dependent magnitude of completeness.
`H`	H-parameter for a Helmstetter model of time-dependent magnitude of completeness.

Table 4: Key/value pairs that appear within the `parameters` object.	
Parameters for the international ETAS statistical model. (see van der Elst *et al*. 2022, https://doi.org/10.1785/0220210222)	
`ams`	ETAS ams-value, which specifies the productivity of the mainshock. This is a maximum likelihood estimate.
`a`	ETAS a-value, which specifies the productivity of aftershocks. This is a maximum likelihood estimate.
`b`	Gutenberg-Richter b-value. This is a fixed value.
`p`	Omori exponent p. This is a maximum likelihood estimate.
`c`	Omori time offset c, in days. This is a maximum likelihood estimate.
Parameters for the domestic ETAS statistical model (preliminary).	
`b`	Gutenberg-Richter b-value. This is a fixed value.
`alpha`	ETAS alpha-value, that specifies how productivity scales with magnitude. It is typically the same as the b-value. This is a fixed value.
`c`	Omori time offset c, in days. It is typically a maximum likelihood estimate, but it could be a fixed value.
`p`	Omori exponent p. It is typically a maximum likelihood estimate, but it could be a fixed value.
`n`	ETAS branching ratio n. The branching ratio effectively determines the productivity of aftershocks. It is typically a maximum likelihood estimate, but it could be a fixed value.
`zams`	Productivity of the mainshock and foreshocks. Usually it is relative to the productivity of aftershocks, *e.g.*, a positive value indicates that the mainshock is more productive than aftershocks after accounting for the difference in magnitude. It is typically a maximum likelihood estimate, but it could be a fixed value.
`zmu`	Background rate, in units of M3 earthquakes per day. It is typically zero, but it could be a fixed positive value, and it could be a maximum likelihood estimate.
`simCount`	The number of ETAS simulations that were run to generate the forecast.
`simDuration`	The duration of each ETAS simulation, in days. It is typically 365 days, which is long enough to generate the 1-year forecast.

Each element in the `forecast` array represents one time period: 1 day, 1 week, 1 month, or 1 year. The key/value pairs appearing in each array element are shown in the following table.

Table 5: Key/value pairs that appear within each element of the `forecast` array.	
`timeStart`	The start of the time period. The value is an integer giving the number of milliseconds since midnight on January 1, 1970. Note that all elements of the `forecast` array have the same start time. Note: The value of `timeStart` is adjusted so that forecasts start at a "round" time, that is, a multiple of 10 minutes or a multiple of 1 hour.
`timeEnd`	The end of the time period. The value is an integer giving the number of milliseconds since midnight on January 1, 1970. At present, it will be either 1 day, 1 week, 1 month, or 1 year after the start time.
`label`	A text string that describes the time period. Currently it is either `"1 Day"`, `"1 Week"`, `"1 Month"`, or `"1 Year"`.
`bins`	An array. Each element of the array corresponds to one aftershock magnitude, and contains the forecast for aftershocks of that magnitude or larger, within the current time period, as described in Table 6.
`aboveMainshockMag`	An object, that contains the forecast for aftershocks larger than the mainshock magnitude, as described in Table 7. In principle, the `aboveMainshockMag` entry is optional, but so far it has always been included.

Each element in the `bins` array represents one aftershock magnitude. It contains the forecast for aftershocks of that magnitude or larger, within the relevant time period. The key/value pairs appearing in each array element are shown in the following table.

Table 6: Key/value pairs that appear within each element of the `bins` array.	
`magnitude`	The magnitude. Almost always, the `bins` array contains elements for magnitudes 3.0, 4.0, 5.0, 6.0, and 7.0. For very large earthquakes, there may also be an element for magnitude 8.0. For some small earthquakes, it is possible that there is no element for magnitude 7.0, and possibly none for magnitude 6.0.
`p95minimum`	An integer, giving the lower limit of the 95% confidence interval.
`p95maximum`	An integer, giving the upper limit of the 95% confidence interval. It is forecasted that there is a 95% probability that the number of aftershocks, with the given magnitude or larger, within the relevant time period, lies between `p95minimum` and `p95maximum` inclusive.
`probability`	A floating-point number. It is the forecasted probability that at least one aftershock, with the given magnitude or larger, occurs within the relevant time period.
`median`	An integer. It is the forecasted median number of aftershocks, with the given magnitude or larger, within the relevant time period. Note: Prior to January 2022, the `median` key/value pair was not included.
`fractileValues`	An array that contains the probability distribution for the forecasted number of aftershocks. Its length equals the length of `fractileProbabilities` (usually 197 elements), and it contains a list of integers that represent numbers of aftershocks. The integers are always in non-decreasing order. Suppose that N is an integer appearing in `fractileValues`, and that P is the corresponding entry of `fractileProbabilities`. Then N is the minimum number of aftershocks, with the given magnitude or larger, within the relevant time period, such that that probability of N or fewer aftershocks is greater than or equal to P. Note: This was added in April 2023. Its addition made `p95minimum`, `p95maximum`, and `median` redundant, however they are still included for compatibility and ease of use.

Table 6: Key/value pairs that appear within each element of the `bins` array.	
`barPercentages`	An array that contains the percentage value for each bar in the bar graph. Its length equals the length of `barLabels`, and it contains a list of integers that represent percentages. (Fractional percentages are not allowed.) It is guaranteed that the first element of this array is the complement of the rounded `probability` value (rounded to the nearest whole percent, except that anything over 99% is rounded to 100%, and anything less than 1% is rounded to 0%). It is not guaranteed that the elements of the array sum to 100%. Note: This was added in November 2023.

The `aboveMainshockMag` object contains the forecast for aftershocks larger than the mainshock, within the relevant time period. The key/value pairs appearing in the object are shown in the following table.

Table 7: Key/value pairs that appear within the `aboveMainshockMag` object.	
`magnitude`	The magnitude of the mainshock
`probability`	A floating-point number. It is the forecasted probability that at least one aftershock, with magnitude larger than the mainshock, occurs within the relevant time period.
`fractileValues`	See description in Table 6.
`barPercentages`	See description in Table 6.

Example of `forecast.json`


```
{
  "creationTime": 1737657861110,
  "expireTime": 1769194800000,
  "advisoryTimeFrame": "1 Month",
  "template": "Mainshock",
  "injectableText": "",
  "observations": [
    {"magnitude": 3.0, "count": 145},
    {"magnitude": 4.0, "count": 14},
    {"magnitude": 5.0, "count": 1},
    {"magnitude": 6.0, "count": 0},
    {"magnitude": 7.0, "count": 0}
  ],
  "model": {
    "name": "Reasenberg-Jones (1989, 1994) aftershock model (Sequence Specific)",
    "reference": "#url",
    "parameters": {
      "a": -2.67,
      "b": 1.0,
      "magMain": 7.0,
      "p": 1.08,
      "c": 0.05,
      "aSigma": 0.0508,
      "pSigma": 0.0616,
      "Mcat": 3.2,
      "F": 1.0,
      "G": 4.5,
      "H": 0.75,
      "regionType": "circle",
      "regionCenterLat": 40.3592,
      "regionCenterLon": -124.8534,
      "regionRadius": 70.0
    }
  },
  "fractileProbabilities": [
    0.01, 0.015, 0.02, 0.025, 0.03, 0.035, 0.04, 0.045, 0.05, 0.055,
    0.06, 0.065, 0.07, 0.075, 0.08, 0.085, 0.09, 0.095, 0.1, 0.105,
    0.11, 0.115, 0.12, 0.125, 0.13, 0.135, 0.14, 0.145, 0.15, 0.155,
    0.16, 0.165, 0.17, 0.175, 0.18, 0.185, 0.19, 0.195, 0.2, 0.205,
    0.21, 0.215, 0.22, 0.225, 0.23, 0.235, 0.24, 0.245, 0.25, 0.255,
    0.26, 0.265, 0.27, 0.275, 0.28, 0.285, 0.29, 0.295, 0.3, 0.305,
    0.31, 0.315, 0.32, 0.325, 0.33, 0.335, 0.34, 0.345, 0.35, 0.355,
    0.36, 0.365, 0.37, 0.375, 0.38, 0.385, 0.39, 0.395, 0.4, 0.405,
    0.41, 0.415, 0.42, 0.425, 0.43, 0.435, 0.44, 0.445, 0.45, 0.455,
    0.46, 0.465, 0.47, 0.475, 0.48, 0.485, 0.49, 0.495, 0.5, 0.505,
    0.51, 0.515, 0.52, 0.525, 0.53, 0.535, 0.54, 0.545, 0.55, 0.555,
    0.56, 0.565, 0.57, 0.575, 0.58, 0.585, 0.59, 0.595, 0.6, 0.605,
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    0.66, 0.665, 0.67, 0.675, 0.68, 0.685, 0.69, 0.695, 0.7, 0.705,
    0.71, 0.715, 0.72, 0.725, 0.73, 0.735, 0.74, 0.745, 0.75, 0.755,
    0.76, 0.765, 0.77, 0.775, 0.78, 0.785, 0.79, 0.795, 0.8, 0.805,
```

```
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  0.91, 0.915, 0.92, 0.925, 0.93, 0.935, 0.94, 0.945, 0.95, 0.955,
  0.96, 0.965, 0.97, 0.975, 0.98, 0.985, 0.99
],
"barLabels": [0, 1, 2, 5, 10, 20, 50, 100, 200, 500],
"forecast": [
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    "timeEnd": 1737745200000,
    "label": "1 Day",
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        "p95maximum": 2,
        "probability": 0.2796,
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        ],
        "barPercentages": [72, 23, 5, 0, 0, 0, 0, 0, 0, 0]
      },
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        "p95maximum": 1,
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        "median": 0,
        "fractileValues": [
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      0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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    ],
    "barPercentages": [100, 0, 0, 0, 0, 0, 0, 0, 0, 0]
  },
  {
    "magnitude": 6.0,
    "p95minimum": 0,
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    "probability": 3.309E-4,
    "median": 0,
    "fractileValues": [
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